Decomposition of micromotion signals into physically interpretable perturbation modes via lattice based simulations

Esteban Acerbo[1,3,4,a], Mariela I. Bellotti[1,2,5], Fabian J. Bonetto[1,2,3,4]

[1] Laboratorio de Cavitación y Biotecnología, Centro Atómico Bariloche (CAB), Bariloche, Argentina.

[2] Comisión Nacional de Energía Atómica (CNEA), Centro Atómico Bariloche (CAB), Bariloche, Argentina

[3] Consejo Nacional de Investigaciones Ciencia y Técnicas (CONICET), Buenos Aires, Argentina.

[4] Universidad Nacional de Cuyo - Instituto Balseiro, Bariloche, Argentina.

[5] Centro Patagónico Interdiciplinario de Investigaciones en Salud (CePIIeS), Bariloche, Argentina.

April 16, 2026

Abstract

Confluent cell monolayers over electrodes exhibit continuous impedance fluctuations that encode morphological and mechanical information at sub-second timescales, driven by lateral displacements, tight-junction remodeling, and biophysical parameter space heterogeneity. Decomposing these fluctuations into their constituent physical mechanisms remains an open problem, as each time step provides only two scalar observables while the underlying state space is high-dimensional. This work addresses the decomposition through physically parameterized lattice-based simulation libraries built on a finite-difference model sharing the Giaever and Keese analytical model parametrization. Four perturbation libraries were computed, representing electrode-monolayer relative movement, subcellular tight-junction geometry modification, collective brick-wall rearrangement, and biophysical parameter microperturbations. Together, these libraries cover the experimental phase space for four physiological states: cell seeding, confluence, electrical wound healing, and post-healing confluence. Two reconstruction algorithms were evaluated: a greedy nearest-prototype method that assigns one mode per library per time step by sequentially minimizing the running residual, and a sparse Viterbi algorithm that enforces smooth state trajectories at the cost of fit quality. Both produce physically interpretable, non-degenerate state sequences for each experimental electrode.

1 Introduction

Understanding individual and collective cell movement provides direct insight into the biochemical and mechanical signals underlying tissue formation, from embryogenesis to organ morphogenesis, and the dysregulation that drives pathological processes including cancer invasion and metastasis [1, 2]. Confluent epithelial monolayers exhibit collective dynamics analogous to liquids approaching a glass transition, displaying spatial heterogeneities [2, 3] and back and forth oscillations [4]. In the field of monolayer monitoring Electric Cell-substrate Impedance Spectroscopy (ECIS) technique provides a continuous, non-invasive readout of the collective state of a cell monolayer adhered to a planar gold electrode by monitoring its spectral impedance. In addition, by monitoring the cell monolayer at sub-second resolution with a fixed excitation frequency, [5, 6, 7, 8] the impedance evolution reveals the collective correlated cell movement or micromotion, as signal fluctuations providing of morphological and toxicity information [9, 10, 11, 12, 13].

The objective of this project is to decompose the impedance experimental micromotion time series into physically interpretable contributions from discrete perturbation modes, producing a state sequence that describes the dominant physical mechanism at each time step. The dataset analyzed here was originally reported in [13] as part of a quantitative fractal characterization of MDCKII monolayer cultures; the present work re-examines those recordings through the lens of the numerical simulation framework extending the Giaever-Keese analytical model. From the original dataset four micromotion time series were selected, all measured at 1 kHz, which proved to be the most sensitive excitation frequency to culture-condition changes.

All signals were adaptively detrended by subtracting a centered moving-average filter with window W from the raw $R(t)$ and $X(t)$ signals. This acts as a high-pass filter, preserving fluctuations at timescales shorter than W while removing slow biological drift. The window is group-adaptive: $W = 900\,\mathrm{s}$ for confluent and post-healing recordings, and $W = 300\,\mathrm{s}$ for seeding and wound-healing recordings, where the impedance rises monotonically over the measurement period and a shorter window is required to track micromotion. The resulting fluctuation signals are stored as $\delta R(t)$ and $\delta X(t)$ for statistical analysis.

[a]Send any correspondence by email to: esteban.acerbo@ib.edu.ar - ORCID: 0000-0001-7021-5208

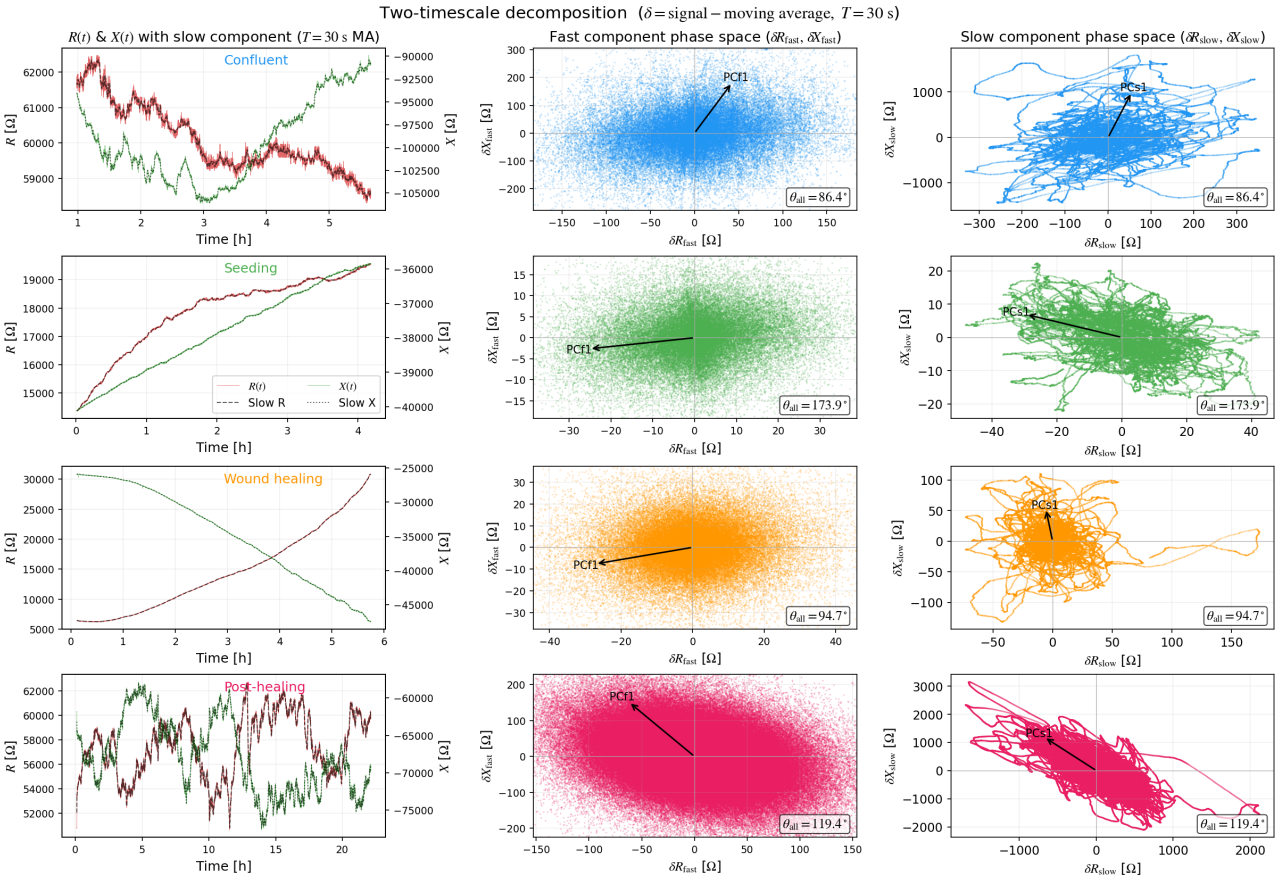


Figure 1: Two-timescale decomposition of the four culture conditions (rows: seeding, confluent, wound healing, post-healing). The raw $R(t)$ and $X(t)$ signals are adaptively detrended to obtain $\delta R(t)$ and $\delta X(t)$; a second moving average ($T_{\mathrm{cut}} = 30\,\mathrm{s}$) separates fast ($< 30\,\mathrm{s}$) and slow ($> 30\,\mathrm{s}$) components. **Left column:** raw $R(t)$ (red) and $X(t)$ (green) time series with the slow component overlaid as dashed lines. **Central column:** fast-component phase space $(\delta R_{\mathrm{fast}}, \delta X_{\mathrm{fast}})$ with PC1 arrow and principal-axis angle θ_{PC}. **Right column:** slow-component phase space $(\delta R_{\mathrm{slow}}, \delta X_{\mathrm{slow}})$ with PC1 arrow and θ_{PC}.

Principal component analysis (PCA) diagonalises the covariance matrix of $(\delta R, \delta X)$, yielding a principal axis angle θ_{all} (measured from the positive δR axis, CCW positive) and standard deviations $s_1 \geq s_2$ along and perpendicular to that axis. A moving-average filter of width $T_{\mathrm{cut}} = 30\,\mathrm{s}$ further decomposes each signal into a fast component ($< 30\,\mathrm{s}$, capturing individual cell micromotion events) and a slow component ($> 30\,\mathrm{s}$, capturing collective reorganization), with PCA applied separately to each, as shown in Fig. 1. For seeding and wound-healing, the fast and slow principal-axis angles are nearly identical, indicating that a single dominant mechanism operates across all timescales. The post-healing recording is the sole exception, exhibiting a clear angular separation between timescales.

To characterize if the fractal scaling is homogeneous across timescales, two time-domain algorithms are therefore applied to decompose and cross-validate the scaling behavior across timescales: the Higuchi [14, 15] algorithm and Rescaled Range (R/S) analysis [16, 17, 18], both described in the previous work [13]. These estimates are obtained from the fully detrended $\delta R(t)$ and δX signals; they are not directly comparable to the values reported which were computed on raw, non-detrended impedance trajectories. Each signal is divided into analysis windows of 12 500 points and for each window the algorithm's log-log scaling curve is computed, obtaining the statistical fractal dimention [19, 20]. For the Higuchi method, the log-log curves of all windows were averaged and a single breakpoint was located by minimising the total piecewise residual sum of squares; the breakpoint was then forced on each individual window to extract fast-timescale (D_f^{fast}) and slow-timescale (D_f^{slow}) slopes separately. R/S was treated with a single slope only, as its lag-range structure does not admit a reliable piecewise decomposition.

The results are summarised in Table 1. For δR, fractal dimension estimates are consistently high across almost all states (Higuchi single-slope range 1.67–1.87). Post-healing yields the lowest values (1.67 ± 0.04 Higuchi, 1.43 ± 0.04 R/S), indicating a measurable reduction in signal complexity following to culture wound closure and radial cell reordening [21], while Seeding and Wound healing are comparable (≈ 1.87 and 1.86 by Higuchi), suggesting similar short-term fluctuation structure in both poorly-confluent and actively remodeling cultures.

The Higuchi breakpoint decomposition of δR consistently reveals a fast segment exceeding the slow segment, indicating that short-timescale dynamics are geometrically more complex. For δX, a sharper contrast between states emerges. Seeding records a near-Brownian dimension of 1.920 ± 0.006, while Confluent and Post-healing drop to 1.43 ± 0.03 and 1.40 ± 0.02 respectively, consistent with reactance fluctuations becoming smoother as the monolayer matures and membrane dynamics become collectively constrained. Wound healing is a notable exception, Higuchi and R/S, both with standard deviations larger than those of any other state or signal combination. This elevated dispersion across windows indicates that the δX fractal scaling is not stationary during the wound-healing phase, likely reflecting the spatial heterogeneity, migrating cells, and intact monolayer coexist and contribute incoherently to the measured reactance fluctuations. The Higuchi breakpoint columns for Wound healing δX should therefore be interpreted with caution, as the forced-split procedure finds an unstable inflection on an already variable averaged curve.

Table 1: Fractal dimension of δR and δX estimated by two algorithms. Values are mean ± standard deviation across analysis windows. Single: full-range slope; Fast: short-timescale segment; Slow: long-timescale segment at forced breakpoint. R/S fast and slow values are omitted as the forced breakpoint is not suited to the R/S scaling structure.

Culture state	δR				δX			
	Higuchi			R/S	Higuchi			R/S
	Single	Fast	Slow	Single	Single	Fast	Slow	Single
Seeding	1.87(3)	1.91(2)	1.77(5)	1.80(2)	1.920(6)	1.950(2)	1.83(2)	1.78(2)
Confluent	1.81(3)	1.85(2)	1.75(6)	1.74(3)	1.43(3)	1.68(5)	1.39(3)	1.43(4)
Wound healing	1.86(3)	1.91(2)	1.80(4)	1.78(3)	1.8(1)	1.92(5)	1.7(1)	1.66(8)
Post-healing	1.67(4)	1.78(3)	1.52(5)	1.43(4)	1.40(2)	1.69(2)	1.33(2)	1.38(3)

2 Bioimpedance Models

Giaever and Keese developed a model to quantify the biophysical properties of a confluent monolayer from its impedance spectrum [5, 22]. The model considers an infinite electrode covered by an infinite, compact layer of identical cylindrical cells obtaining the specific impedance of the electrode covered by a confluent monolayer as described in Eq. 1.

$$\frac{1}{Z_{\text{cov}}} = \frac{1}{Z_n(Z_n + Z_m)} \left[Z_n + \frac{Z_m}{\frac{\gamma r_c}{2}\frac{I_0(\gamma r_c)}{I_1(\gamma r_c)} + R_b\left(\frac{1}{Z_n} + \frac{1}{Z_m}\right)} \right]. \tag{1}$$

With I_i are the i-th modified Bessel function of first kind, Z_n the specific impedance of the bare electrode, $Z_m(C_m)$ the membrane-specific impedance, R_b the specific tight-junction resistance and r_c the cell radius. For experimental data fitting it is convenient to define $\alpha = r_c\sqrt{\rho/h}$, where ρ is the medium resistivity and h the cleft height. Consequently, the specific impedance of the cell-covered electrode is characterized by three parameters, α, R_b, and C_m.

The specific impedance of the bare electrode Z_n can be modelled as a constant-phase element (CPE) as describe in Eq. 2. The electrode geometry and electrolyte interface introduces a constriction resistance in series with each electrode. Because R_{constr} is constant in time, it cancels in the fluctuation signal $\delta Z = Z(t) - \langle Z \rangle$ and does not affect the reconstruction of cellular micromotion.

$$Z_{\text{CPE}} = \frac{1}{A\,(j\omega)^n} + R_{\text{constr}}, \tag{2}$$

The simulation engine used in this work is a numerical finite-difference model based on the Giaever-Keese current balance approach, accounting for the electrode size and enabling the simulation of asymmetric cultures in fixed states snapshots. The simulated culture confluent construction over the electrode can be seen in Fig 2 with the spectral impedance estimation, compared to experimental confluent data and the Giaever and Keese estimation using the same input parameters. The circular electrode of diameter $350\,\mu$m (96W1E+ PET Applied Biophysics CultureWare) is centered on the domain. The culture is composed of square cells divided into sub-cellular square-pixel membrane tiles with a shared one-pixel-wide tight-junction ring at cell boundaries. For each square pixel the lateral and apical voltage differences are calculated enabling the passage of current from one cell cleft space in to the neighbors, this passage presents an impedance proportional to α. Each pixel is assigned an apical impedance resistive ($\propto R_b$) or capacitive ($\propto C_m$) depending on being tight junction or interior membrane. The finite-difference voltage balance is solved for each pixel at each frequency, from which the electrode impedance is integrated. The parameters used throughout are $\alpha = 19.4 \pm 1.1$, $R_b = (14.4 \pm 2.0)\,\Omega\text{cm}^2$, and $C_m = (4.95 \pm 0.12)\,\mu\text{F/cm}^2$, obtained from a spectral fit to a confluent MDCKII monolayer using Eq. 1.

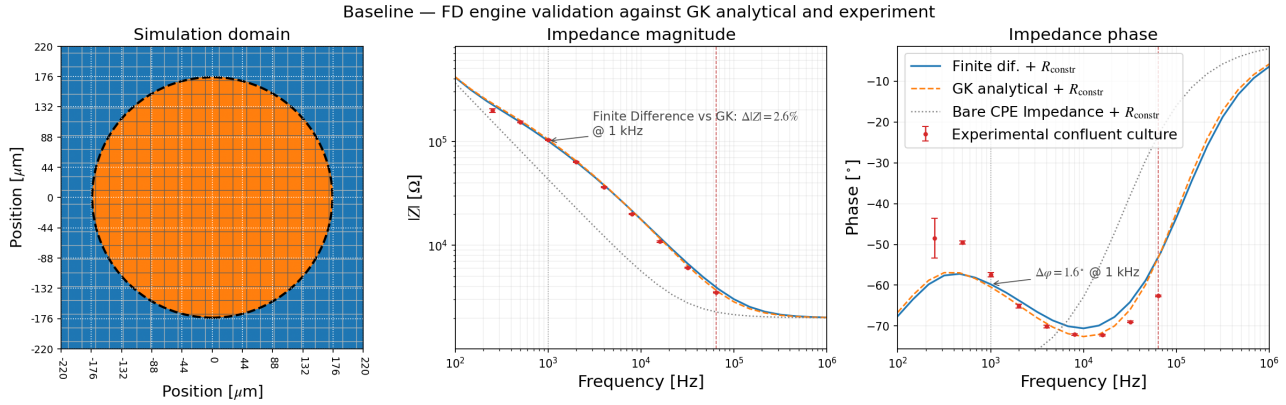


Figure 2: Finite-difference simulation construction and spectral impedance estimation. **Left:** Simulation domain showing the confluent square-cell culture map over the circular electrode. **Center:** Impedance modulus $|Z|(f)$ and **Right:** phase angle $\angle Z(f)$; each panel overlays the finite-difference result, the Giaever-Keese analytical model (Eq. 1) evaluated with the same parameters, experimental confluent MDCKII data, and the bare electrode modeled as a CPE. The 96W1E+ PET Applied Biophysics CultureWare consist in two identical electrodes. The total cultureware impedance result in the sum of two GK monolayers plus a constriction resistance or two finite difference simulations plus the constriction resistance. Each simulation or model estimation is based in half the bare electrode total impedance.

The simulation framework is built on a fixed spatial lattice in which cell identities are assigned to pixel positions rather than tracked as individual mobile agents. This design rests on two complementary principles. The impedance measured at the electrode is determined by the aggregate spatial distribution of membrane, junction, and bare substrate pixels; permutation of two cells across the lattice leaves the impedance spectrum invariant, so tracking individual cell trajectories would expand the sampled hyperspace without adding observational content. Moreover, each impedance evaluation requires a full sparse finite-difference solve of the electrode boundary-value problem, making continuous-dynamics simulation computationally prohibitive at the required frequency resolution; the library approach replaces trajectory integration with a finite set of static equilibrium geometries whose impedance responses are pre-computed once and reused during reconstruction.

3 Perturbation Libraries

Four such libraries were computed by systematically modifying the confluent construction described above, each spanning a specific physical mechanism. Each mode stores the full impedance spectrum over 25 log-spaced frequencies from 100 Hz to 1 MHz, and the $(\delta R, \delta X)$ values are computed as the impedance deviation from the confluent baseline (Fig. 2). The libraries stand for independent additive small perturbations, taking the confluent baseline to the micromotion specific state, covering the experimental phasespace.

The first implemented simulations, Library A represents a rigid translation of the monolayer grid over the electrode disc to represent culture oscillations. As the monolayer shifts, the mapping of tight-junction and membrane pixels onto the electrode integration domain changes while the electrode geometry and CPE parameters remain fixed. Simulations span a lower-triangular matrix sweep $\Delta x \in [0, 40]$ px, $\Delta y \in [0, \Delta x]$, exploiting the reflection symmetry of a square cell on a circular electrode to avoid redundant modes. The perturbation direction at 1 kHz is consistently $(+\delta R, -\delta X)$, as seen in Fig. 3.

Library B models the systematic widening of cell junctions driven by intercellular mechanical tension, in which cells deform collectively along a common direction. Each cell simultaneously converts an L-shaped group of membrane pixels in one corner into tight-junction (R_b) type; the modification occupies l_w pixels along both the horizontal and vertical arms, ranging from one pixel to the full cell side. The circular symmetry of the electrode makes the choice of corner unimportant. A coupling parameter rescales R_b in proportion to the arm length l_w, linking the geometric modification to the tight-junction impedance. This coupling between geometry and R_b generates a 2D fan of directions in the $(\delta R, \delta X)$ plane.

Library C models a collective geometric rearrangement associated with glassy monolayer dynamics, in which alternating cell columns are shifted vertically by a pixel offset. This transforms the square packing geometry toward a brick-wall pattern without altering individual cell properties. The perturbation direction at 1 kHz is consistently $(+\delta R, +\delta X)$, making it the only library that provides a positive contribution in both impedance components.

Library D represents perturbations that produce impedance fluctuations without any geometric rearrangement.

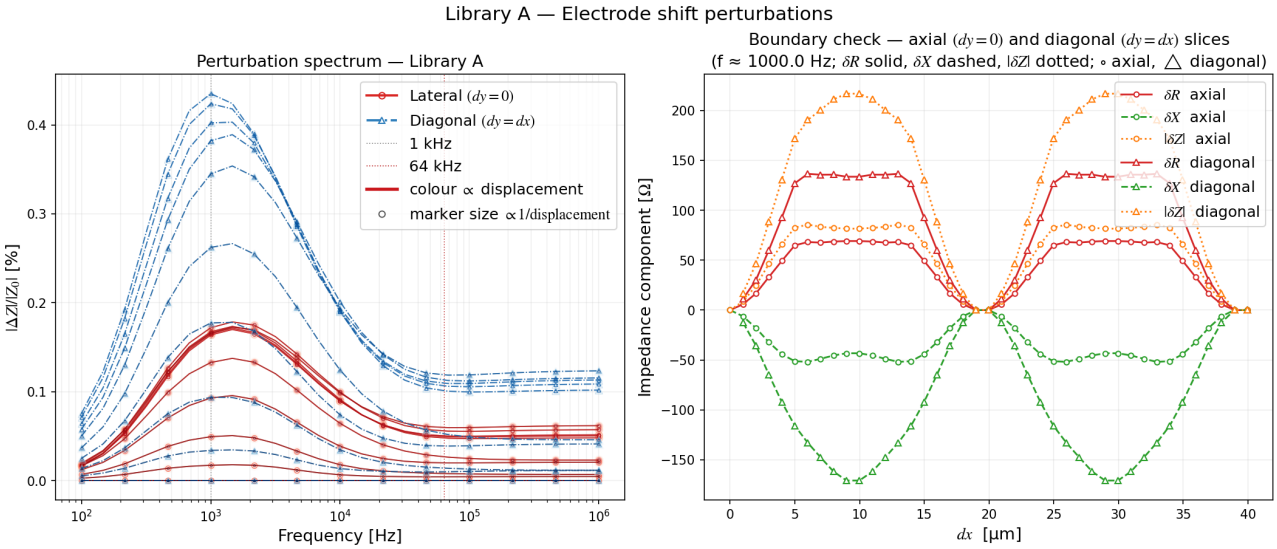


Figure 3: Library A electrode-shift perturbations. **Left:** Normalized perturbation magnitude $|\Delta Z|/|Z_0|$ (%) as a function of frequency for the lateral-shift and diagonal-shift series; line color and marker size increase with displacement magnitude. **Right:** Real part δR, imaginary part δX, and magnitude $|\delta Z|$ at 1 kHz as a function of lateral or diagonal displacement dx (µm), shown as separate overlaid curves for the lateral and diagonal series.

Changes in the cell–substrate cleft geometry or paracellular barrier properties translate directly into variations of α, R_b, and C_m. The library is constructed from the Cartesian product of fractional perturbations in $[-1\%, +1\%]$ for each of the three parameters, and the resulting modes cover all four quadrants of the $(\delta R, \delta X)$ plane as shown in Fig. 6.

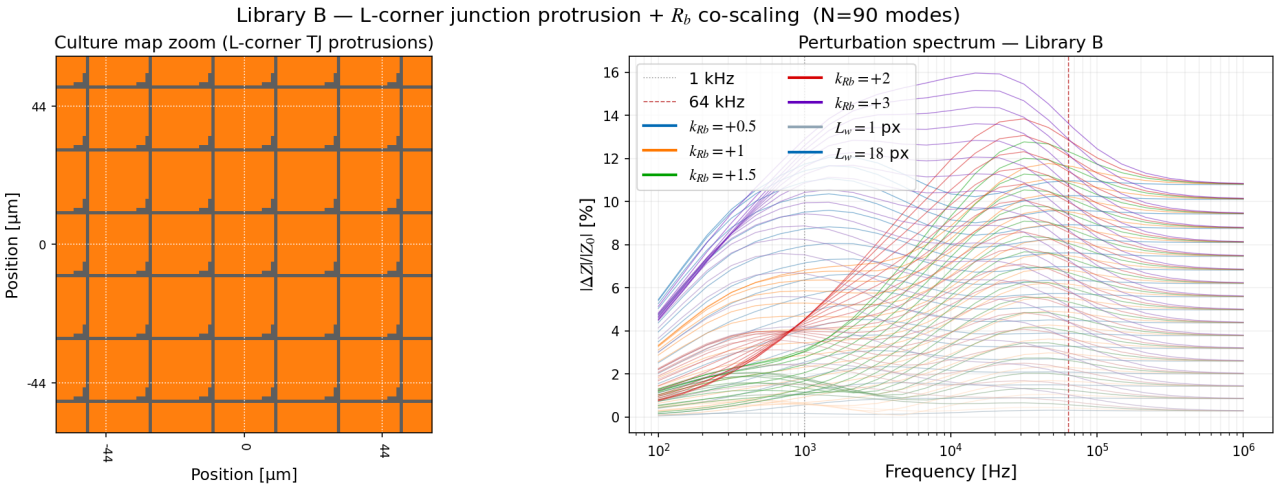


Figure 4: Library B: L-corner junction opening with R_b co-perturbation. **Left:** Simulation domain zoom showing the L-shaped tight-junction modification in the lower-right corner of each cell, parameterized by arm length l_w and coupling parameter K_{RB}. **Right:** Perturbation amplitude spectrum $|\delta Z|/|Z_0|$ (%) as a function of frequency for all Library B modes, spanning the full range of l_w and K_{RB}.

The combined set of library modes must be capable of recreating the experimentally observed micromotion fluctuations. To verify this, the Minkowski sum of the four libraries was computed and its convex hull overlaid on the experimental confluent trajectory in Fig. 7. The hull delimits the region of the $(\delta R, \delta X)$ plane accessible to the reconstruction algorithm, although it does not reflect the density of accessible points within that region. It encloses 95.9% of the confluent experimental trajectory; the remaining events lie outside the current parameter range of the libraries, not outside the physical model itself.

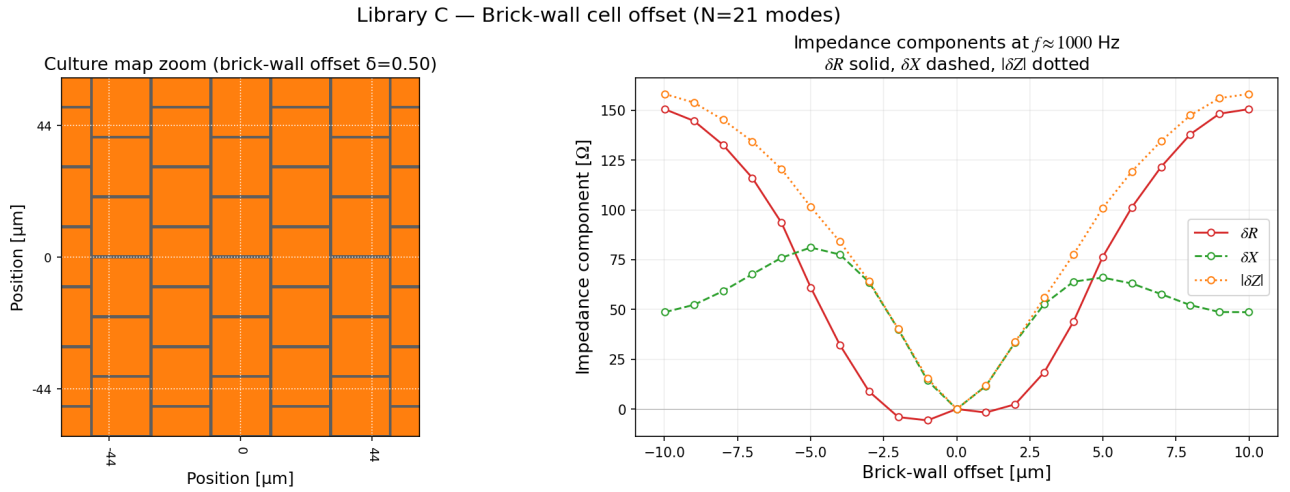


Figure 5: Library C: collective geometric rearrangement from square packing geometry toward a brick-wall pattern without altering individual cell properties. **Left:** Simulation domain zoom showing maximum brick-wall pattern. **Right:** Real part δR, imaginary part δX, and magnitude $|\delta Z|$ at 1 kHz as a function of vertical displacement.

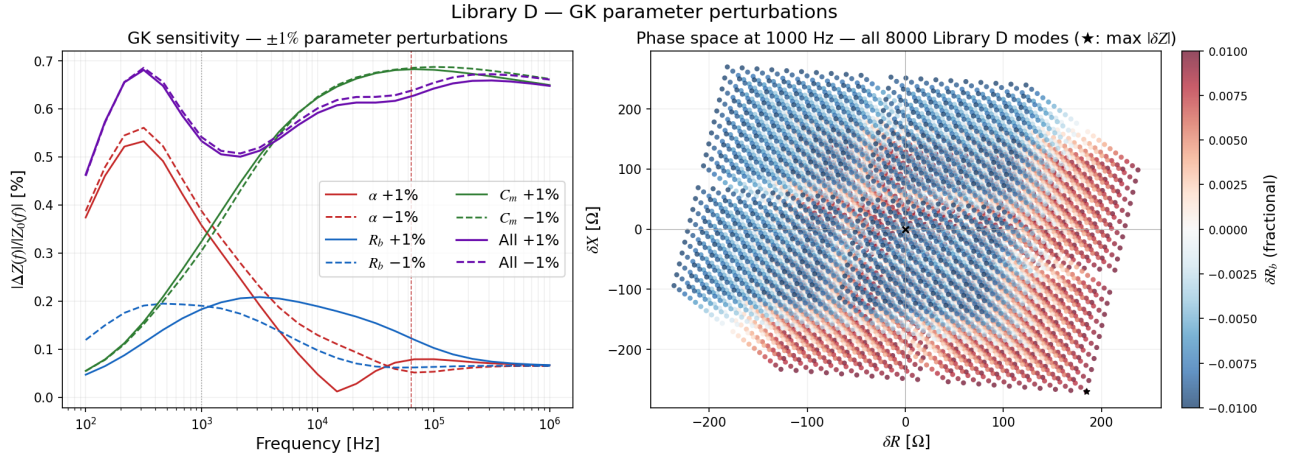


Figure 6: Library D, biophysical parameter microperturbation. **Left:** Individual and combined parameter sensitivity curves. **Right:** Phase-space scatter $(\delta R, \delta X)$ of all Library D modes at 1 kHz.

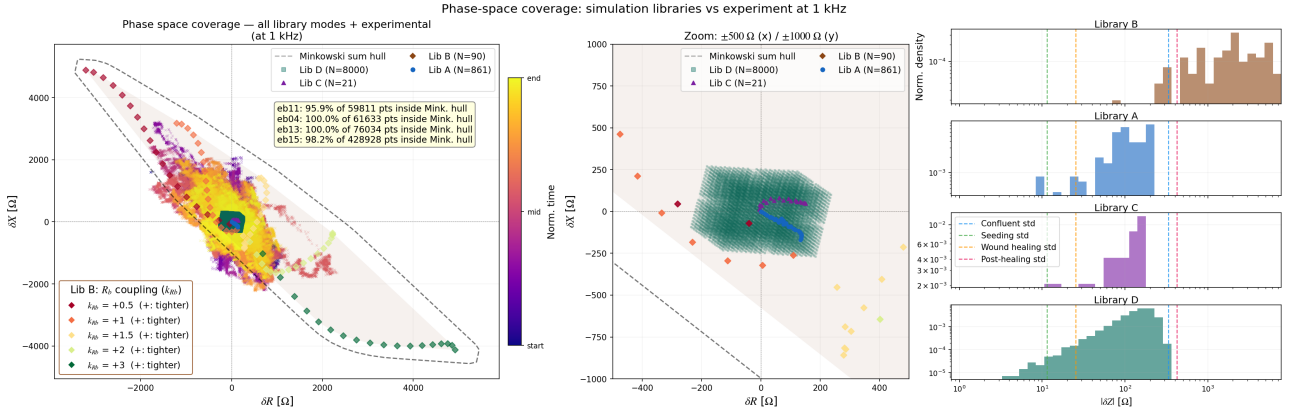


Figure 7: Phase-space coverage. **Left:** Scatter $(\delta R, \delta X)$ of all individual library modes (Libraries A, B, C, D shown with distinct markers and colors) overlaid with the experimental confluent trajectory. **Center:** Zoom into the low-amplitude region showing Libraries A, C, and D. **Right:** Cumulative distributions of $|\delta Z|$ amplitude, shown separately for each library (one panel per library); vertical dashed lines mark the experimental standard deviations $\sigma_{\delta R}$ and $\sigma_{\delta X}$ projected onto the amplitude axis.

With the libraries defined, the experimental signal $\mathbf{y}(t) = (\delta R(t),\, \delta X(t))^\top \in \mathbb{R}^2$ is decomposed at each time step by a greedy nearest-prototype algorithm. The four libraries are processed in a fixed sequence ordered by decreasing perturbation amplitude: B first (largest amplitude and dominant phase-space direction), then D, A, and finally C. At each stage the single mode from the current library that minimises the Euclidean distance to the running residual is selected and its δZ vector subtracted; after all four libraries have been processed, the remainder $\boldsymbol{\epsilon}(t)$ is the reconstruction residual. Quality is reported as $\mathrm{RMSE}_{\delta R}$ and $\mathrm{RMSE}_{\delta X}$, the root-mean-square of ϵ_R and ϵ_X across all time steps. The reconstruction assigns four labelled physical states per time step, one from each library, constituting a physically interpretable, non-degenerate description of the monolayer condition that forms a multivariate time series amenable to further analysis.

As an alternative, a sparse Viterbi algorithm was implemented to enforce temporal continuity in the state sequence [23]. The outer sequential structure is identical to the greedy pipeline: the four libraries are processed in the same fixed order on sequential residuals. Within each library pass, however, the per-step nearest-prototype search is replaced by a full dynamic-programming solve over the entire time series, finding the globally optimal state sequence for that library given the current residual (Eq. 3–4). Before each Viterbi pass a null mode ($\delta Z = 0$) is prepended to the library, allowing the algorithm to express that the library contributes nothing at time steps where the residual is already small. The observation cost is the squared Euclidean distance $C_{\mathrm{obs}}(t, j) = \|\mathbf{r}(t) - \delta\mathbf{Z}_j\|^2$, and the transition cost $T(i, j)$ in Eq. 4 is zero when the distance between adjacent modes does not exceed a threshold Δ_L and infinite otherwise, hard-forbidding large jumps. The distance d_L is library-specific: Euclidean in pixel-displacement space for Library A; Euclidean in normalised $(K_{Rb},\, l_w)$ space for Library B; Chebyshev distance in grid-index space for Library D (to handle the non-uniform parameter grid); and absolute index difference for Library C.

Replace with:

$$\{s^*(t)\}_{t=0}^{T-1} = \underset{s(t)\in\mathcal{M}_L}{\arg\min} \sum_{t=0}^{T-1} C_{\mathrm{obs}}(t,\, s(t)) + \sum_{t=1}^{T-1} T\big(s(t{-}1),\, s(t)\big), \tag{3}$$

$$T(i,\, j) = \begin{cases} 0 & \text{if } d_L(i, j) \le \Delta_L, \\ \infty & \text{otherwise,} \end{cases} \tag{4}$$

The libraries were computed for confluent cultures only; the numerical model could in principle simulate partial coverage, but this was deliberately excluded to avoid the combinatorial expansion introduced by randomized cell arrangements on partially covered electrodes. For non-confluent recordings, a calibration factor $cal = |Z_{0,\mathrm{exp}}|/|Z_{0,\mathrm{sim}}|$ is therefore applied as a uniform scalar multiplier to all library mode vectors before reconstruction, correcting for the impedance level offset between the confluent simulation baseline and the experimental state.

4 Results

Figure 8 shows the greedy and Viterbi reconstructions of the confluent culture micromotion. Both algorithms track the δX fluctuation closely, while a systematic δR residual persists, consistent with the libraries' dominant phase-space direction lying closer to the δX axis. The greedy reconstruction achieves component-wise RMSE of 54 Ω (δR) and 49 Ω (δX), corresponding to 43% and 9.4% of the respective experimental standard deviations. The Viterbi regularization increases both values, and its residual exhibits spikes where the temporal-continuity constraint prevents the algorithm from following rapid experimental excursions.

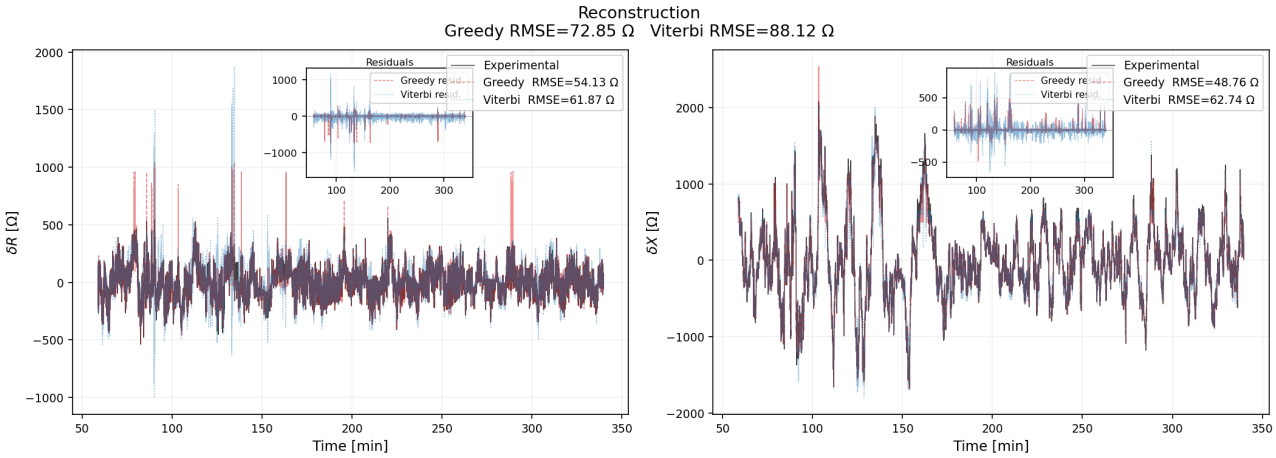


Figure 8: Reconstruction of impedance fluctuations for the confluent culture. **Main panels:** $\delta R(t)$ and $\delta X(t)$ as a function of time for the experimental signal (black), the greedy reconstruction (red), and the Viterbi reconstruction (blue). **Insets:** Corresponding residuals $\epsilon_R(t)$ and $\epsilon_X(t)$; $\mathrm{RMSE}_{\delta R}$ and $\mathrm{RMSE}_{\delta X}$ for each algorithm are indicated in the legend.

Table 2 summarizes reconstruction quality for all four experimental cases. The confluent and post-healing electrodes were reconstructed without calibration. Seeding and wound-healing required $cal < 1$, which rescales the library vectors to the lower experimental impedance level, increasing the density of accessible states in the $(\delta R, \delta X)$ plane. This rescaling reduces the greedy RMSE to below 6% of $\sigma_{\delta R}$ and below 12% of $\sigma_{\delta X}$ for both non-confluent cases, indicating that the perturbation libraries capture the dominant fluctuation mechanisms once the impedance baseline is matched.

Table 2: Reconstruction RMSE for four focus electrodes. Greedy: nearest-prototype; Viterbi: temporal proximity regularization. RMSE in Ω; % relative to experimental standard deviation of each component.

		Greedy				Viterbi			
Culture state	cal	δR [Ω]	% σ	δX [Ω]	% σ	δR [Ω]	% σ	δX [Ω]	% σ
Seeding	(0.42)	1.16	5.8	1.10	11.4	3.4	17	2.6	27.0
Confluent	OFF	54	43	49	9.4	62	49.2	63	12.1
Wound healing	(0.38)	1.46	5.9	1.07	3.5	5	20.5	3.6	11.7
Post-healing	OFF	36.7	9.2	25	4.0	55	13.5	40	6.4

Figure 9 summarizes the per-library state sequences selected by each algorithm for the confluent electrode. The left column shows the statistical distribution of each parameter over the full recording, providing a global view of which regions of the library phase space are accessed. The central and right columns show the temporal evolution of the same parameters within a 5-min window centered on the peak $|\delta R_{\mathrm{exp}}|$ event.

The (K_{Rb}, l_w) scatter (Library B) shows that both algorithms sample the same discrete set of configurations throughout the recording. This is consistent with Fig. 7, since these modes provide the coarse correction for confluent cultures. Library D parameter distributions are not statistically different between the two algorithms: $\delta\alpha$ is skewed toward -1%, δR_b shows a broad distribution across mid-range values with pronounced peaks at the perturbation extremes, and δC_m is centered around zero. Moreover, the Viterbi zoom reveals an anticorrelation between $\delta\alpha$ and the other two parameters, indicating that the reconstruction consistently opposes the sign of the cleft-geometry perturbation to those of the junction-resistance and membrane-capacitance perturbations. The displacement distribution (Library A) is markedly different between the two algorithms. The greedy distribution is multimodal, reflecting the algorithm's freedom to jump between discrete regions of the shift library at each time step. The Viterbi solution is dominated by a single population near zero displacement, with occasional selection of the library boundary or its symmetric counterpart at the first negative displacement values. This contrast reflects the temporal-continuity penalty against large jumps, which forces Viterbi to choose one stable regime and remain near it. The brick-wall offset (Library C) distribution is dominated by few recurrent modes. These narrow occupancy are a direct consequence of Library A and C prominent phase space direction and being processed last in the fitting sequence.

The temporal zoom panels make the algorithmic contrast explicit: the greedy traces switch rapidly between library modes at every time step, achieving the per-step optimum, while the Viterbi traces exhibit piecewise-

constant plateaus separated by discrete transitions. Neither algorithm produces physically implausible states, as both are constrained to the pre-computed library modes.

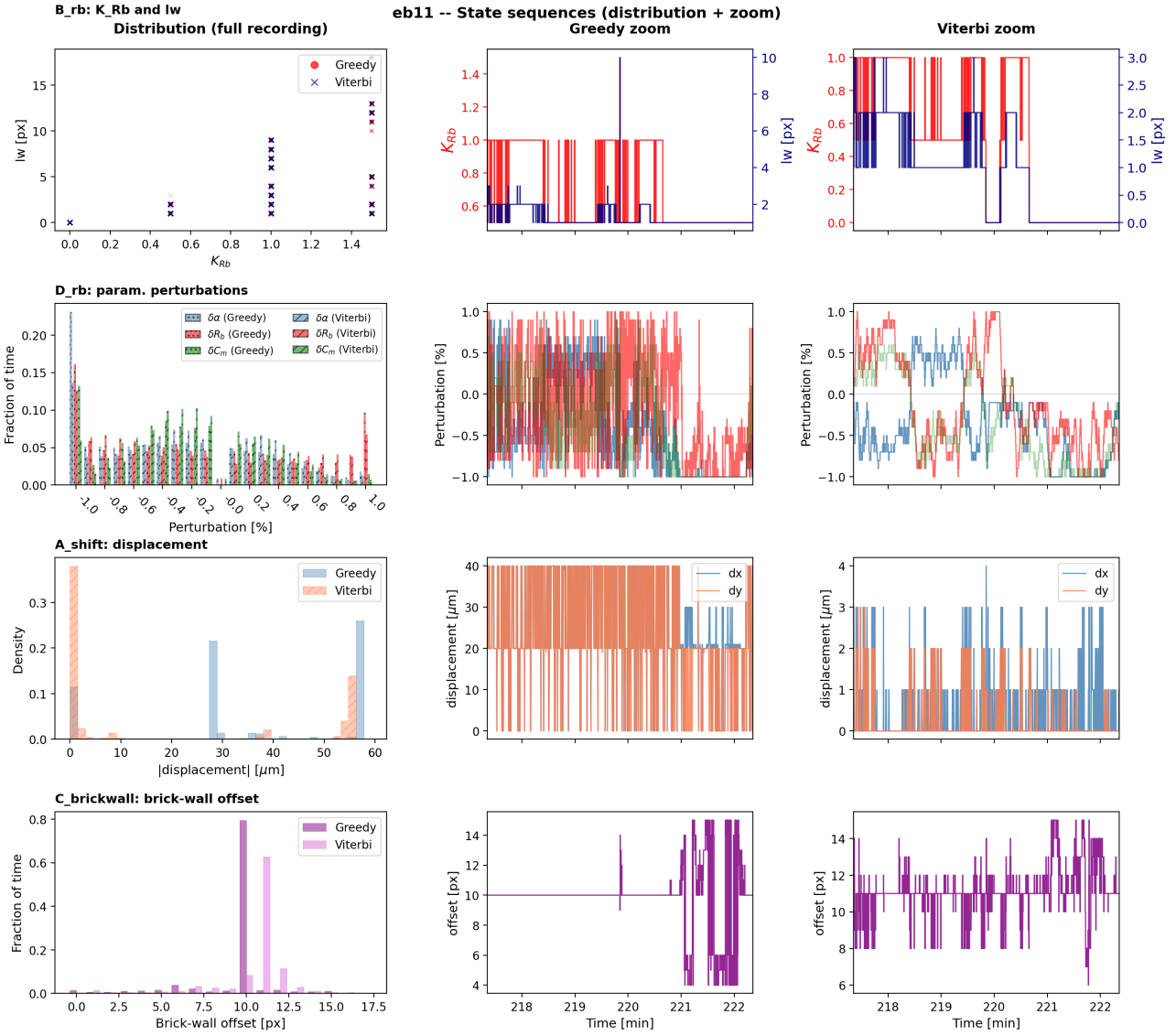


Figure 9: Per-library state sequences for the confluent electrode. **Left column:** distribution of each library parameter over the full recording. Rows from top to bottom: Library B (K_{Rb}, l_w) mode occupancy scatter; Library D fractional perturbations $\delta\alpha$, δR_b, δC_m; Library A displacement magnitude $|d|$; Library C brick-wall offset frequency. **Center and right columns:** temporal evolution of the same parameters within a 5-min window centered on the peak $|\delta R_{\text{exp}}|$ event, for the greedy and Viterbi algorithms respectively.

To validate that the discrete mode representation preserves the temporal correlation structure of the signal, the fractal dimension of each reconstructed series was compared to its experimental counterpart. Table 3 shows agreement within windowed uncertainty for all culture states and both algorithms under the Higuchi estimator. The largest departure appears in the post-healing Viterbi case under R/S estimation ($d_f = 1.59(3)$ vs. experimental 1.43(4)), attributable to the smoothing imposed by the temporal-proximity constraint, which suppresses short-timescale roughness and inflates the apparent fractal dimension.

Table 3: Fractal dimension single-slope d_f — experimental and reconstructed signals (δR and δX). Values are mean with windowed standard deviation in parentheses.

Culture state	δR / δX					
	Higuchi			R/S		
	Exp.	Greedy	Viterbi	Exp.	Greedy	Viterbi
Seeding	1.87(3)/1.920(6)	1.88(3)/1.920(6)	1.88(3)/1.921(8)	1.80(2)/1.78(2)	1.79(2)/1.78(2)	1.77(2)/1.78(2)
Confluent	1.81(3)/1.43(3)	1.81(4)/1.45(3)	1.81(3)/1.48(3)	1.74(3)/1.43(4)	1.74(2)/1.47(3)	1.74(4)/1.53(2)
Wound healing	1.86(3)/1.8(1)	1.86(3)/1.8(1)	1.85(4)/1.8(1)	1.78(3)/1.66(8)	1.79(3)/1.66(8)	1.76(3)/1.66(7)
Post-healing	1.67(4)/1.40(2)	1.67(4)/1.41(2)	1.71(3)/1.46(2)	1.43(4)/1.38(3)	1.44(4)/1.39(3)	1.59(3)/1.47(3)

5 Conclusions

The pipeline demonstrated here produces a physically interpretable, non-degenerate reconstruction of ECIS micromotion time series from four perturbation libraries. The greedy nearest-prototype algorithm achieves the minimum available RMSE and identifies the dominant physical mechanisms at each time step without coefficient degeneracy. The Viterbi temporal-proximity regularization enforces smoother state trajectories at the cost of fit quality. The fractal dimension of the reconstructed signals agrees with that of the experimental signals within windowed uncertainty for all culture states and both algorithms, establishing that the discrete mode representation preserves the temporal correlation structure of the micromotion signal.

The algorithmic contrast between greedy and Viterbi decompositions reflects a fundamental trade-off between fit fidelity and temporal coherence. The greedy algorithm treats each time step as independent, producing the per-step optimum but allowing rapid mode switching that may not correspond to physical state transitions. The Viterbi regularization imposes a physically motivated constraint, since real cells do not jump between arbitrary configurations at each measurement interval. However, the hard-constraint implementation used here systematically increases RMSE, indicating that the underlying experimental signal contains genuine high-frequency content that the regularization cannot accommodate.

The per-library state distributions provide a direct window into the dominant physical mechanisms driving micromotion in the confluent culture. Libraries B and D (tight-junction geometry and biophysical parameter drift) are the principal contributors to the reconstruction, sampled uniformly across the recording and providing the coarse correction that absorbs most of the fluctuation amplitude. Libraries A and C (lateral displacement and brick-wall rearrangement) operate on the residual left by the earlier stages and contribute with narrow occupancy concentrated at small number of recurrent modes.

An unexpected result of the Library D mode evolution is the Viterbi's anticorrelation between $\delta\alpha$ and (δR_b, δC_m) at short timescales, suggesting that cleft-geometry and junction-resistance perturbations move in opposite directions within the selected mode trajectory. This combined signature is consistent with cytoskeletal contractions that tighten the cell-substrate cleft while simultaneously stiffening paracellular junctions, though the broad δR_b distribution indicates that the junction response is not confined to a single value. Verifying whether this asymmetry persists across more culture states and pharmacological perturbations is a natural direction for follow-up work.

The calibration factor required for non-confluent recordings ($cal < 1$ for seeding and wound-healing) exposes a limitation of the current library design. Because the libraries were computed for a fully confluent baseline, partially covered electrodes are represented by rescaling the perturbation amplitudes rather than by modifying the underlying geometry. This approximation is self-consistent within the linear-perturbation regime but does not capture the qualitatively different micromotion dynamics expected at low confluence, where individual cells behave as isolated particles rather than coupled elements of a collective. Extending the library construction to include partially covered electrode configurations would remove this approximation and allow direct reconstruction of the seeding and wound-healing phases without amplitude calibration.

The sensitivity of the ECIS measurement to micromotion perturbations is frequency-dependent through the GK dispersion relations. Analysis of the library sensitivity curves shows that the perturbation amplitude $|\delta Z(f)|$ peaks for most libraries at frequencies below 2 kHz. Above this transition frequency, the measurement is dominated by the CPE electrode response and constriction resistance, resulting in decreasing sensitivity to cellular perturbations. The 1 kHz excitation used in the analyzed dataset therefore sits near the sensitivity maximum, which contributes to the high quality of the reconstruction but also implies that moving to higher frequencies would not improve the signal-to-noise ratio of the micromotion component. Nevertheless, to further improve the precision and reach of this pipeline, future measurements require acquisition protocols that alternate between culture spectral impedance and micromotion monitoring, since each library mode has a precomputed spectral impedance response. If the experimental system provides impedance measurements at multiple frequencies, the search can be conducted in a higher-dimensional observation space, substantially restricting the set of admissible

modes and resolving the phase-space overlap that limits single-frequency reconstruction. Furthermore, spectral shape information provides a physically motivated constraint on which library modes can be consistent with a given experimental state, enabling mode pre-filtering before the search and improving physical interpretability.

Acknowledgements

This work was carried out at the Laboratorio de Cavitación y Biotecnología, Centro Atómico Bariloche, Comisión Nacional de Energía Atómica (CNEA).

Data accessibility

The experimental impedance data and simulation library files supporting this study are available from the corresponding author upon reasonable request.

Declaration of competing interests

The authors declare no competing interests.

References

[1] Xavier Trepat, Michael R Wasserman, Thomas E Angelini, Emil Millet, David A Weitz, James P Butler, and Jeffrey J Fredberg. Physical forces during collective cell migration. *Nature physics*, 5(6):426–430, 2009.

[2] Daniel A Matoz-Fernandez, Kirsten Martens, Rastko Sknepnek, Jean-Louis Barrat, and Silke Henkes. Cell division and death inhibit glassy behaviour of confluent tissues. *Soft matter*, 13(17):3205–3212, 2017.

[3] Silke Henkes, Kaja Kostanjevec, J Martin Collinson, Rastko Sknepnek, and Eric Bertin. Dense active matter model of motion patterns in confluent cell monolayers. *Nature communications*, 11(1):1405, 2020.

[4] Vanni Petrolli, Magali Le Goff, Monika Tadrous, Kirsten Martens, Cédric Allier, Ondrej Mandula, Lionel Hervé, Silke Henkes, Rastko Sknepnek, Thomas Boudou, et al. Confinement-induced transition between wavelike collective cell migration modes. *Physical review letters*, 122(16):168101, 2019.

[5] I. Giaever and C.R. Keese. Micromotion of mammalian cells measured electrically. *Proceedings of the National Academy of Sciences*, 88(17):7896–7900, 1991.

[6] I. Giaever and C.R. Keese. A morphological biosensor for mammalian cells. *Nature*, 366(6455):591–592, 1993.

[7] C.R. Keese and I. Giaever. A biosensor that monitors cell morphology with electrical fields. *IEEE Engineering in Medicine and Biology Magazine*, 13(3):402–408, 1994.

[8] Fabián Eduardo Giana. *Desarrollo de una técnica basada en mediciones de impedancia eléctrica para la discriminación in vitro entre células cancerosas y no cancerosas*. PhD thesis, Universidad Nacional de Uncyo, Instituto Balseiro - CONICET, 2018.

[9] I. Giaever and C.R. Keese. Fractal motion of mammalian cells. *Physica D: Nonlinear Phenomena*, 38(1-3):128–133, 1989.

[10] Douglas C Lovelady, Jennifer Friedman, Sonali Patel, David A Rabson, and Chun-Min Lo. Detecting effects of low levels of cytochalasin b in 3t3 fibroblast cultures by analysis of electrical noise obtained from cellular micromotion. *Biosensors and Bioelectronics*, 24(7):2250–2254, 2009.

[11] Marco Tarantola, Anna Pietuch, David Schneider, Jan Rother, Eva Sunnick, Christina Rosman, Sebastien Pierrat, Carsten Sönnichsen, Joachim Wegener, and Andreas Janshoff. Toxicity of gold-nanoparticles: synergistic effects of shape and surface functionalization on micromotility of epithelial cells. *Nanotoxicology*, 5(2):254–268, 2011.

[12] COLIN S Izzard and LINDA R Lochner. Formation of cell-to-substrate contacts during fibroblast motility: an interference-reflexion study. *Journal of cell science*, 42(1):81–116, 1980.

[13] Esteban Acerbo, Mariela I. Bellotti, and Fabián J. Bonetto. Determining mammalian cells state by fractal micromotion. *Heliyon*, 10(5), 2024.

[14] Tomoyuki Higuchi. Approach to an irregular time series on the basis of the fractal theory. *Physica D: Nonlinear Phenomena*, 31(2):277–283, 1988.

[15] Sou Nobukawa, Teruya Yamanishi, Haruhiko Nishimura, Yuji Wada, Mitsuru Kikuchi, and Tetsuya Takahashi. Atypical temporal-scale-specific fractal changes in alzheimer's disease eeg and their relevance to cognitive decline. *Cognitive Neurodynamics*, 13(1):1–11, 2019.

[16] Jens Feder. *The fractal dimension*. Springer, 1988.

[17] EH Lloyd. *Long-Term Storage: An Experimental Study*. Wiley Online Library, 1966.

[18] Dmitry Mottl. hurst. https://github.com/Mottl/hurst, 2019.

[19] J Guyonnet, E Agoritsas, P Paruch, and S Bustingorry. Statistics of roughness for fluctuating interfaces: A survey of different scaling analyses. *arXiv preprint arXiv:1904.11726*, 2019.

[20] Joe Brewer and Larry Di Girolamo. Limitations of fractal dimension estimation algorithms with implications for cloud studies. *Atmospheric Research*, 82(1-2):433–454, 2006.

[21] W Gamal, S Borooah, S Smith, I Underwood, V Srsen, S Chandran, PO Bagnaninchi, and B Dhillon. Real-time quantitative monitoring of hipsc-based model of macular degeneration on electric cell-substrate impedance sensing microelectrodes. *Biosensors and Bioelectronics*, 71:445–455, 2015.

[22] I. Giaever and C.R. Keese. Monitoring fibroblast behavior in tissue culture with an applied electric field. *Proceedings of the National Academy of Sciences*, 81(12):3761–3764, 1984.

[23] G David Forney. The viterbi algorithm. *Proceedings of the IEEE*, 61(3):268–278, 2005.